ASSIGNMENT : 13

ADVANCED PROGRAMMING

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ROLLNO-CSB24045

SESSSION-SPRING YEAR-2024-28

EXPLORER

C
- assignment13.c
- assignment13.exe
- assignment14.py

OUTLINE
TIMELINE
VS CODE PETS

assignment13.c × | assignment14.py

assignment13.c > main()

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

typedef struct {
    char *data;
    size_t length;
    size_t capacity;
} StringBuffer;

StringBuffer* sb_init(size_t initial_capacity) {
    StringBuffer *sb = (StringBuffer*)malloc(sizeof(StringBuffer));

    if (sb == NULL) {
        printf("Memory allocation failed\n");
        return NULL;
    }

    sb->data = (char*)malloc(initial_capacity * sizeof(char));

    if (sb->data == NULL) {
        printf("Buffer allocation failed\n");
        free(sb);
        return NULL;
    }

    sb->length = 0;
    sb->capacity = initial_capacity;
    sb->data[0] = '\0';

    return sb;
}

void sb_append(StringBuffer *sb, const char *str) {
    size_t str_len = strlen(str);

    while (sb->length + str_len + 1 > sb->capacity) {
```

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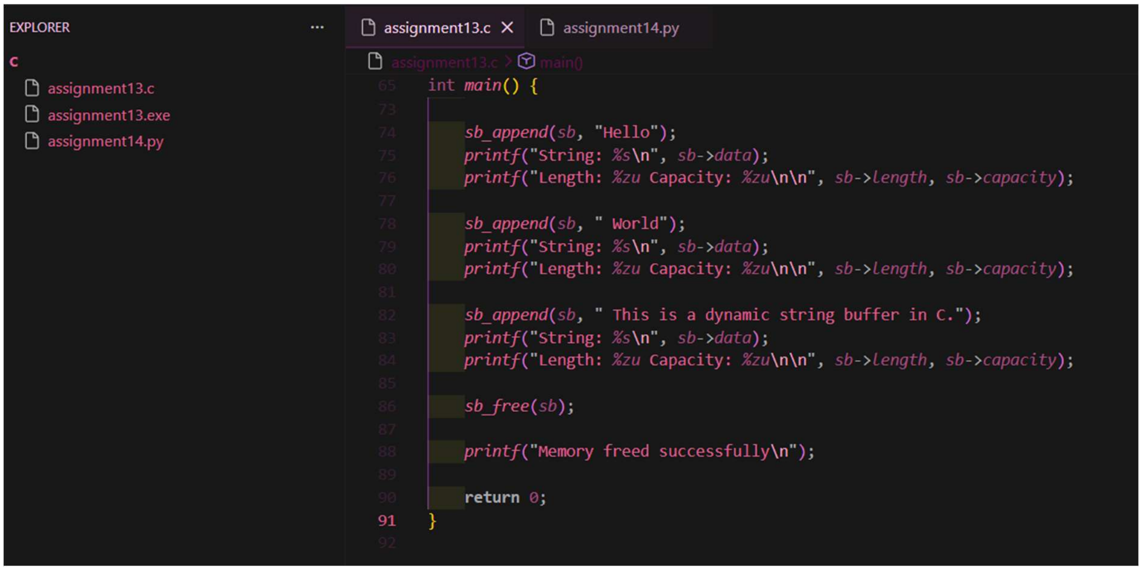
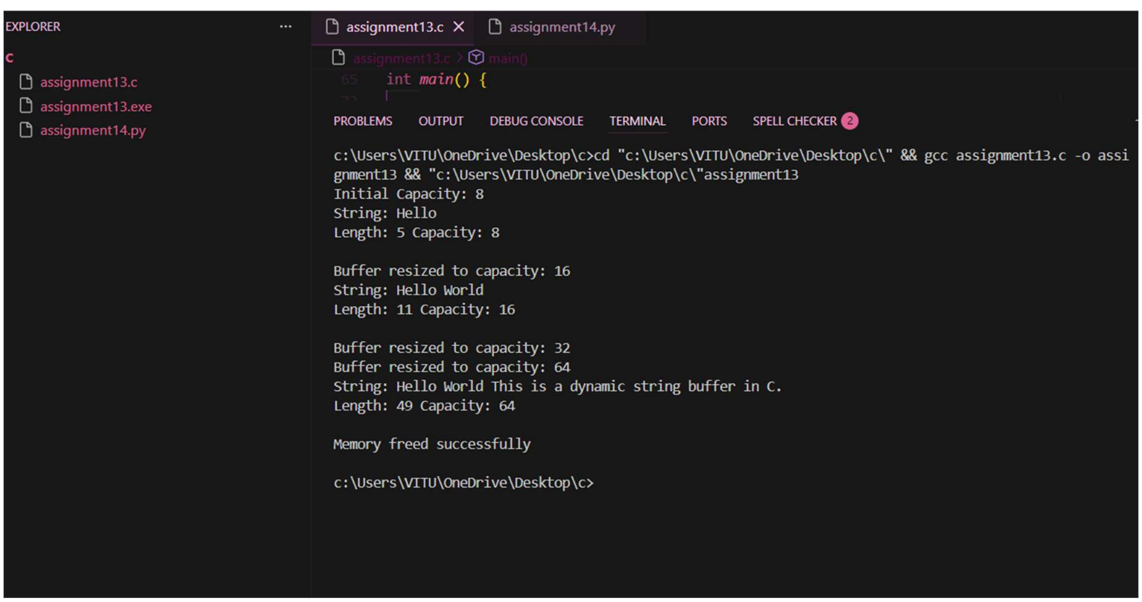
```c
void sb_append(StringBuffer *sb, const char *str) {
    while (sb->length + str_len + 1 > sb->capacity) {
        size_t new_capacity = sb->capacity * 2;

        char *temp = (char*)realloc(sb->data, new_capacity);

        if (temp == NULL) {
            printf("Reallocation failed\n");
            return;
        }

        sb->data = temp;
        sb->capacity = new_capacity;

        printf("Buffer resized to capacity: %zu\n", sb->capacity);
    }

    strcat(sb->data, str);

    sb->length += str_len;
}

void sb_free(StringBuffer *sb) {
    if (sb != NULL) {
        free(sb->data);
        free(sb);
    }
}

int main() {
    StringBuffer *sb = sb_init(8);

    if (sb == NULL) {
        return 1;
    }

    printf("Initial Capacity: %zu\n", sb->capacity);
```


EXPLORER
C
assignment13.c
assignment13.exe
assignment14.py
assignment13.c
assignment14.py
assignment13.c > main()
int main() {
sb_append(sb, "Hello");
printf("String: %s\n", sb->data);
printf("Length: %zu Capacity: %zu\n\n", sb->length, sb->capacity);
sb_append(sb, " World");
printf("String: %s\n", sb->data);
printf("Length: %zu Capacity: %zu\n\n", sb->length, sb->capacity);
sb_append(sb, " This is a dynamic string buffer in C.");
printf("String: %s\n", sb->data);
printf("Length: %zu Capacity: %zu\n\n", sb->length, sb->capacity);
sb_free(sb);
printf("Memory freed successfully\n");
return 0;
}

OUTPUT-


EXPLORER
C
assignment13.c
assignment13.exe
assignment14.py
assignment13.c
assignment14.py
assignment13.c > main()
int main() {
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
SPELL CHECKER 2
c:\Users\VITU\OneDrive\Desktop\c>cd "c:\Users\VITU\OneDrive\Desktop\c\" && gcc assignment13.c -o assignment13 && "c:\Users\VITU\OneDrive\Desktop\c\"assignment13
Initial Capacity: 8
String: Hello
Length: 5 Capacity: 8
Buffer resized to capacity: 16
String: Hello World
Length: 11 Capacity: 16
Buffer resized to capacity: 32
Buffer resized to capacity: 64
String: Hello World This is a dynamic string buffer in C.
Length: 49 Capacity: 64
Memory freed successfully
c:\Users\VITU\OneDrive\Desktop\c>